# Lab 8 - LAN Setup with DHCP using Cisco Packet Tracer

## Aim

To create and configure a Local Area Network (LAN) using one router, one switch, one server, and multiple PCs in Cisco Packet Tracer, and to implement dynamic IP address allocation using a DHCP server. The router is configured to act as the default gateway.

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## Components Used

- 1 × Cisco Router (2911)
- 1 × Cisco Switch (2960-24TT)
- 1 × Server
- 4 × PCs
- Copper straight-through cables

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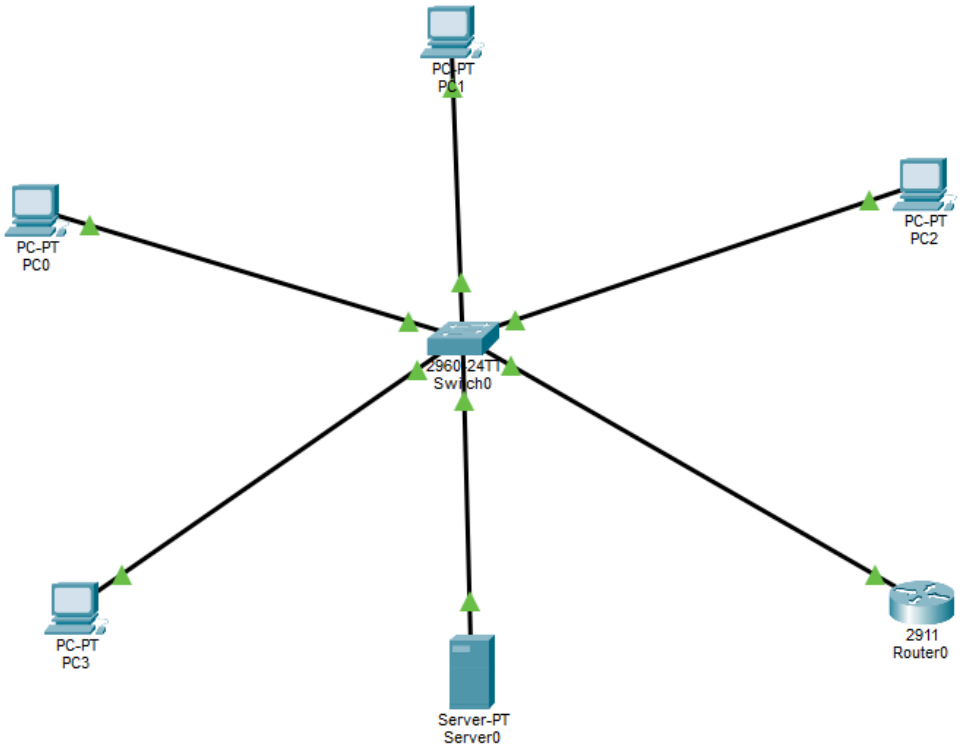
## Network Topology

The following topology was designed where all devices are connected through a central switch:

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# IP Addressing Scheme

- Network: `192.168.1.0/24`
- Default Gateway: `192.168.1.1` (Router)
- Server (Static): `192.168.1.100`
- DHCP Range: Starting from `192.168.1.10`

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# Procedure

## 1. Physical Setup

- All PCs and the server were connected to the switch.
- The switch was connected to the router.
- Proper cabling (Copper Straight-Through) was used.

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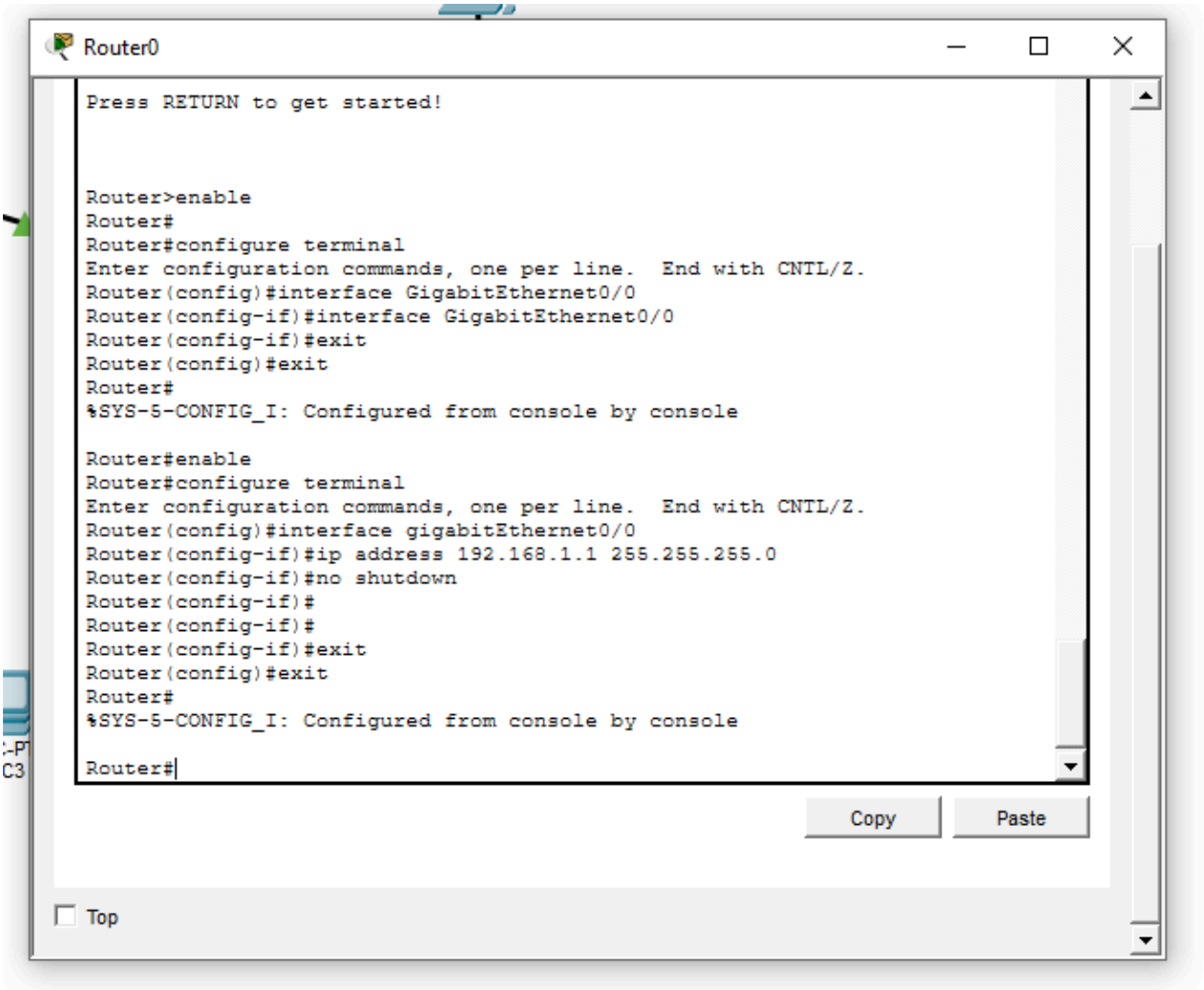
## 2. Router Configuration

The router was configured to act as the default gateway.

### CLI Commands:

```
enable
configure terminal
interface gigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

## Router CLI Output:

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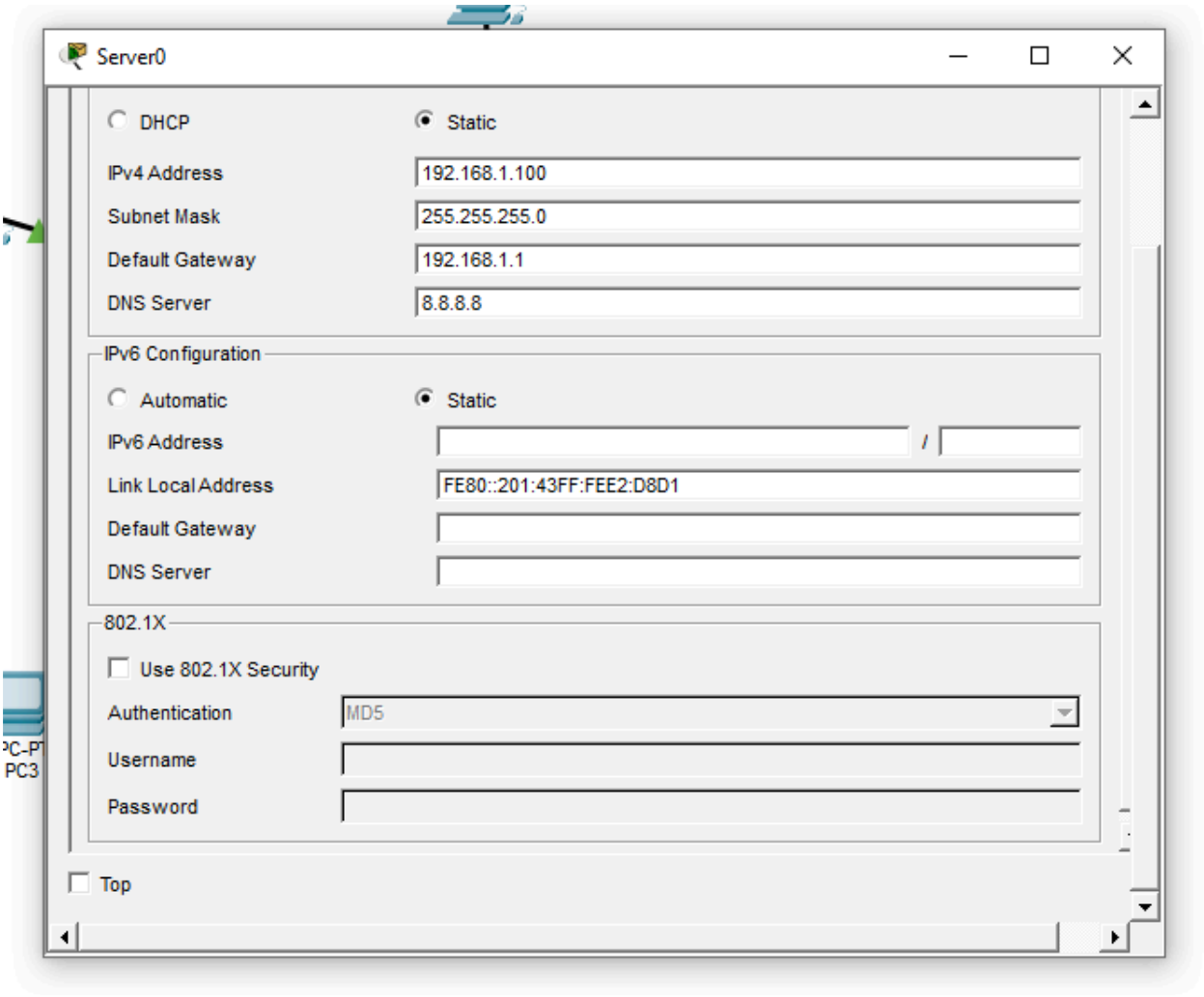
# 3. Server Configuration

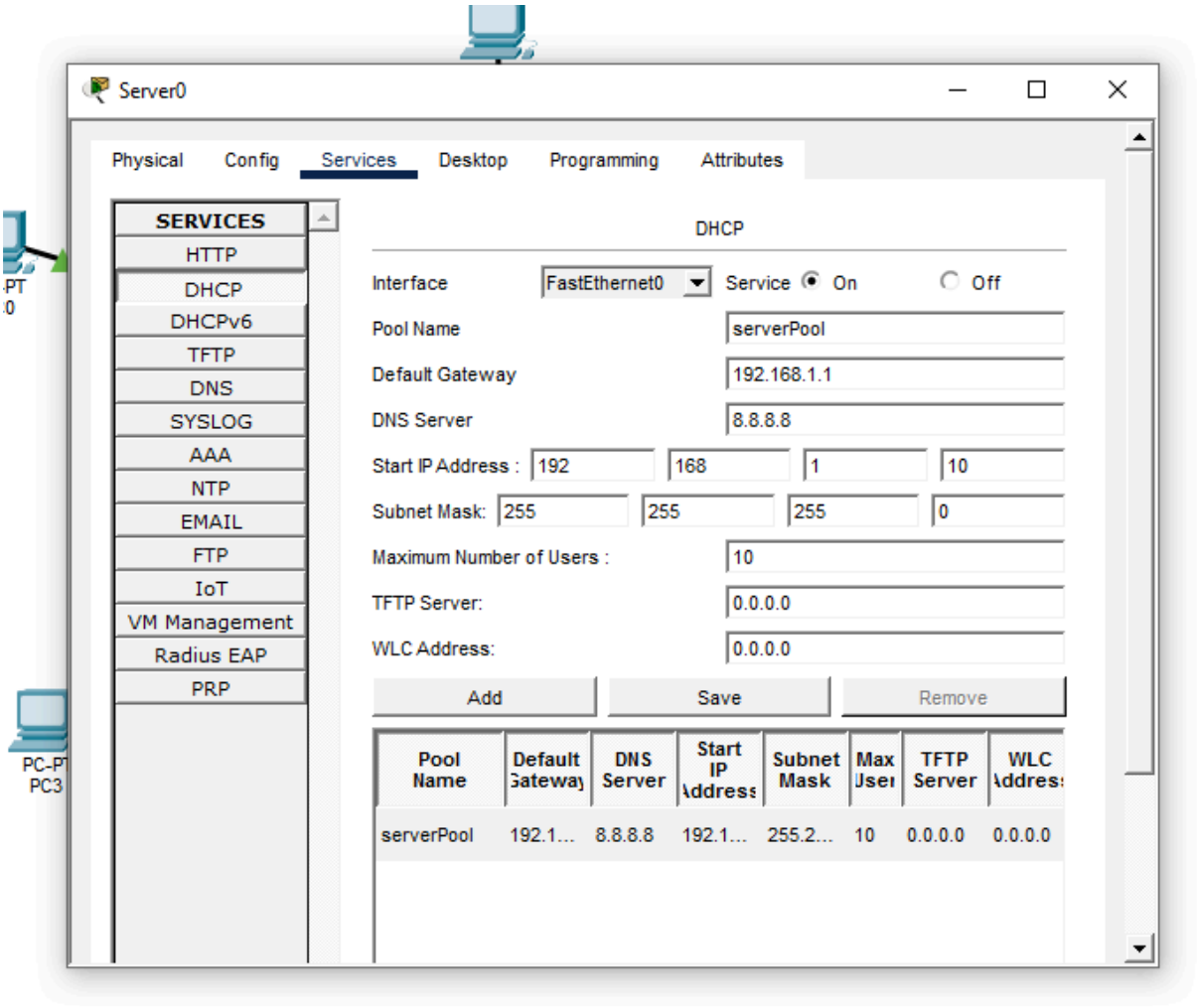
## Static IP Configuration:

- IP Address: `192.168.1.100`
- Subnet Mask: `255.255.255.0`
- Default Gateway: `192.168.1.1`
- DNS Server: `8.8.8.8` (Google)




## DHCP Service Configuration:

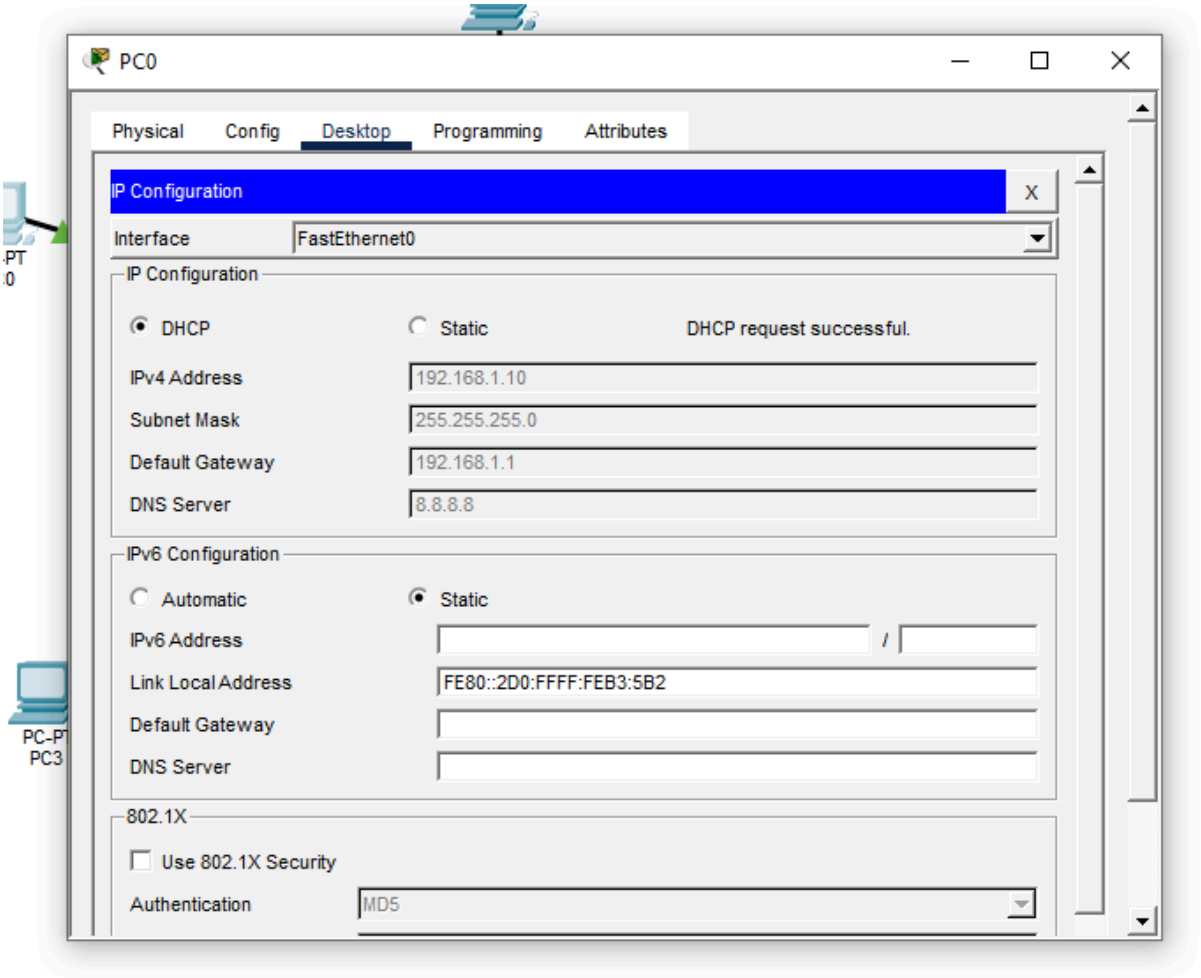
- DHCP Service: Enabled
- Pool Name: `serverPool`
- Default Gateway: `192.168.1.1`
- DNS Server: `8.8.8.8`
- Start IP Address: `192.168.1.10`
- Maximum Users: `10`



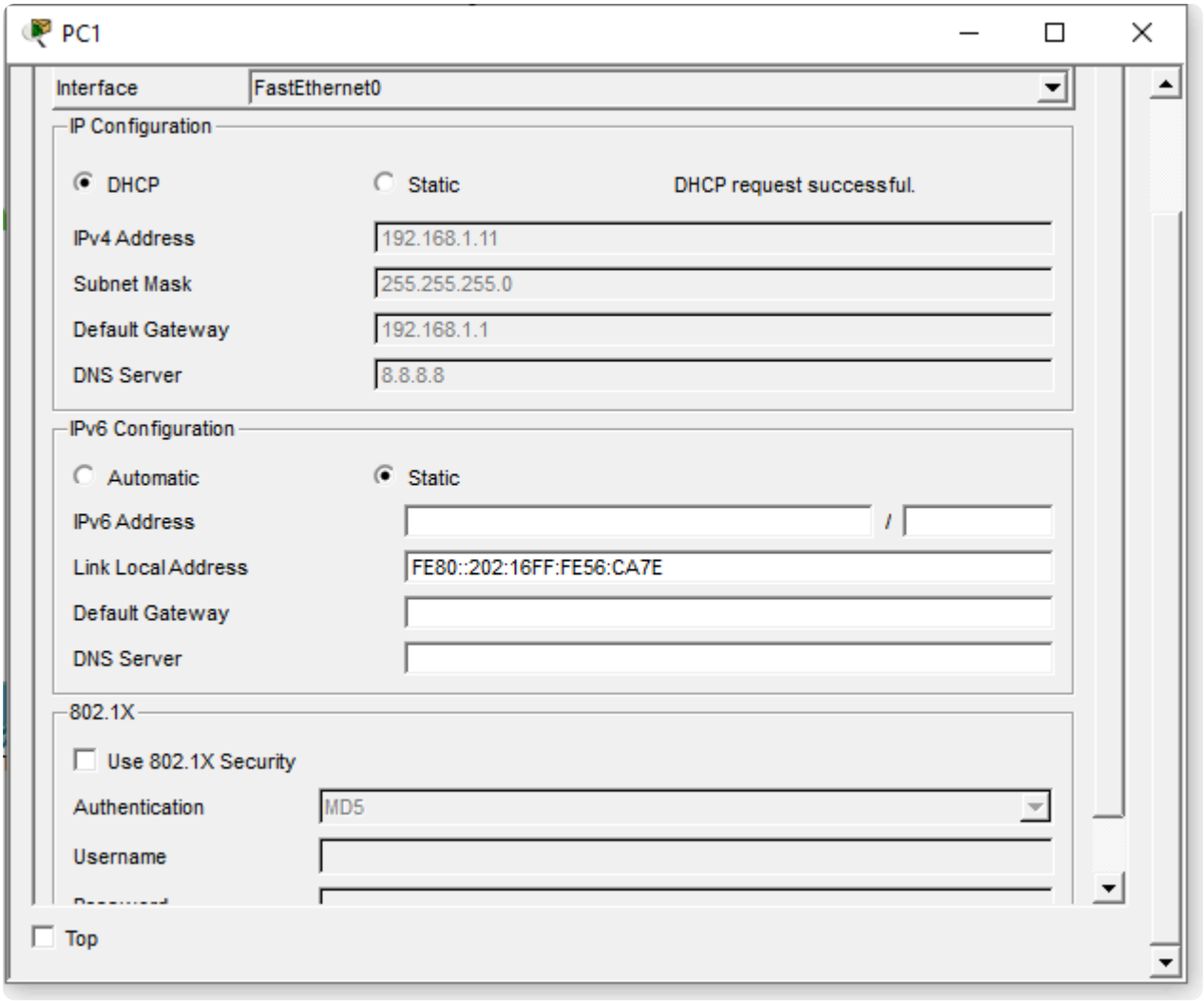

## 4. PC Configuration (DHCP Enabled)

All PCs were set to obtain IP addresses automatically via DHCP.

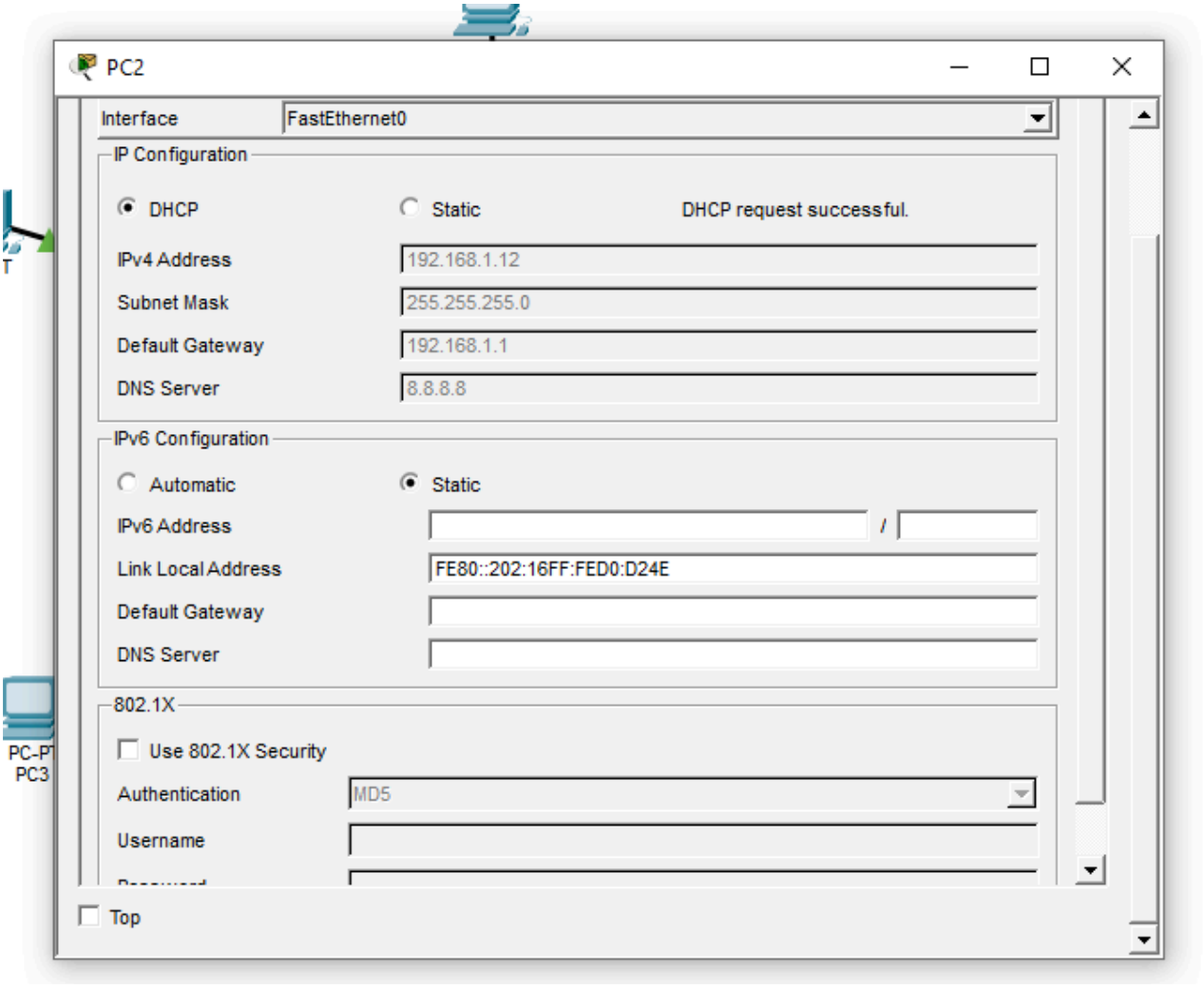
# PC0


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:FFFF:FEB3:5B2
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
PC-PT
PC3

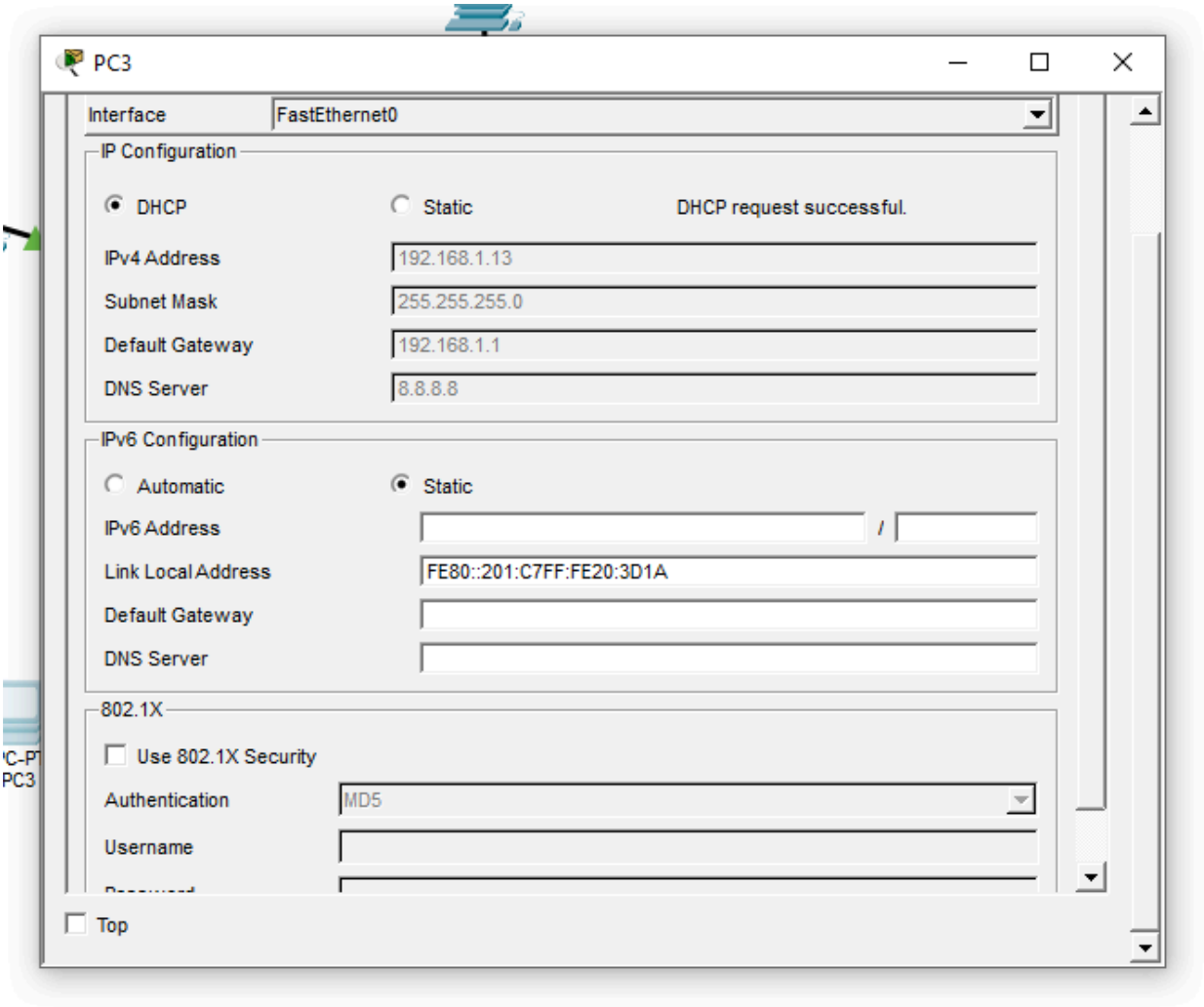
## PC1


PC1
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.1.11
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::202:16FF:FE56:CA7E
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Top

## PC2


PC2
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.1.12
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::202:16FF:FED0:D24E
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Top
PC-PT
PC3

### PC3




- Each PC successfully received:
  - Unique IP address (192.168.1.x)
  - Subnet mask
  - Default gateway
  - DNS server
- DHCP request status shows **"successful"** on all PCs.

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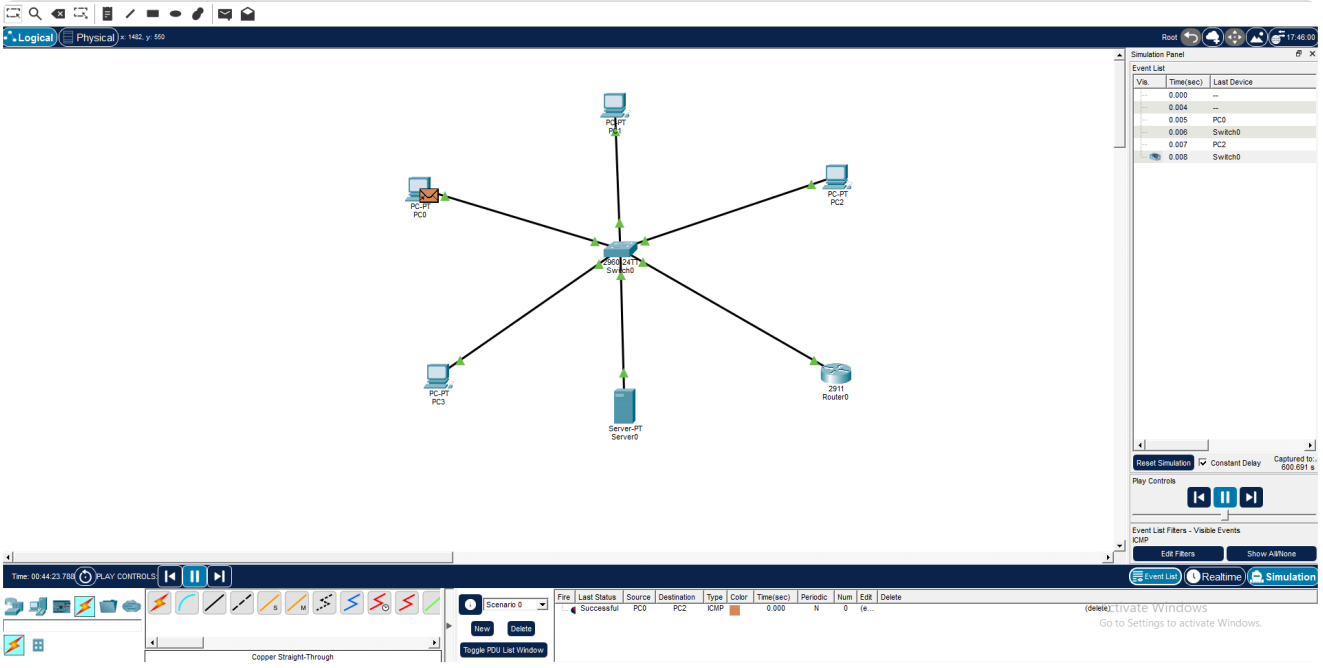
## 5. DHCP Process Verification (Simulation Mode)

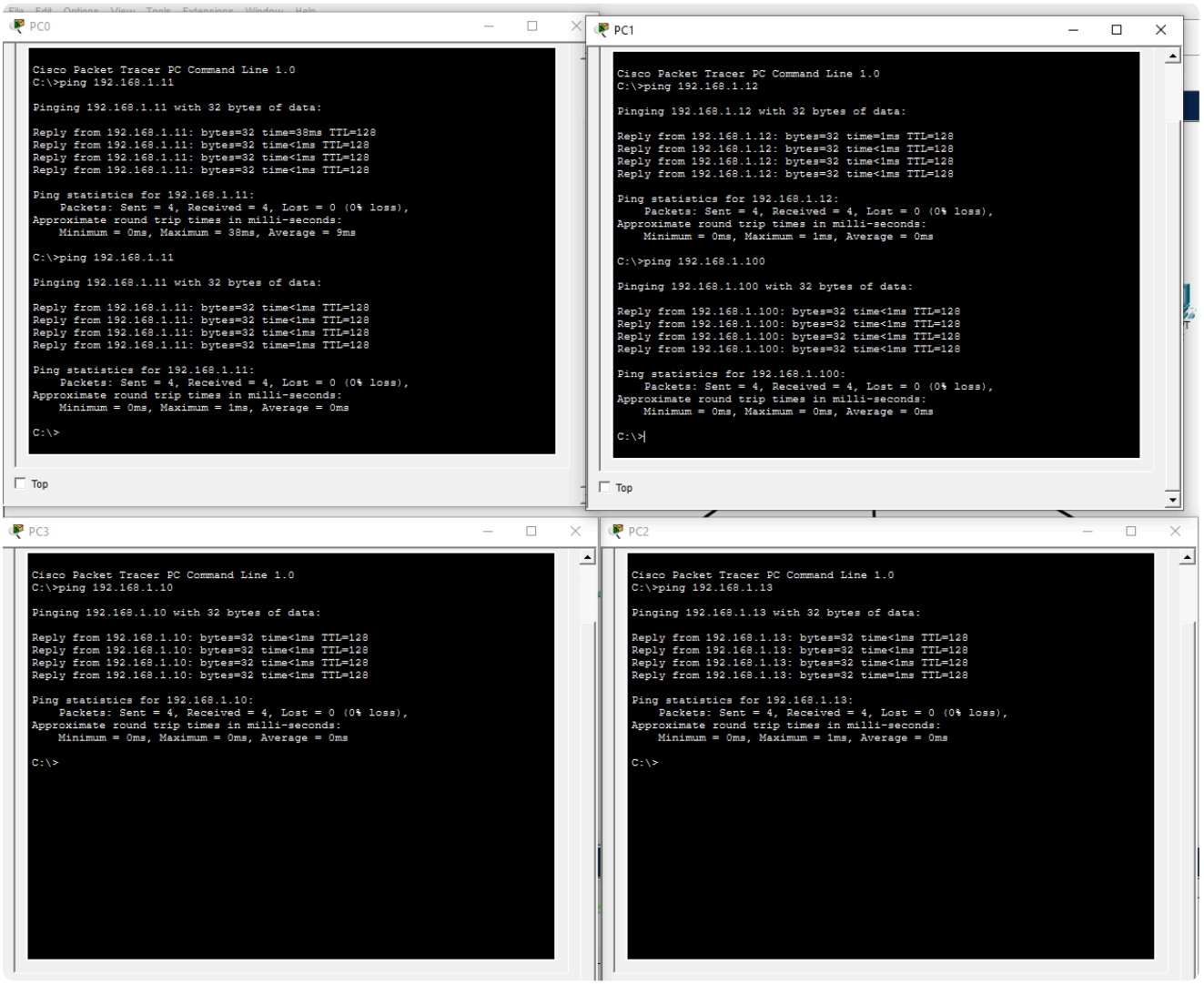
Simulation mode was used to observe the movement of packets from one device to another and also to verify that the network is up and running.




# 6. Connectivity Testing

## PC-to-PC Communication

All PCs were able to successfully ping each other.

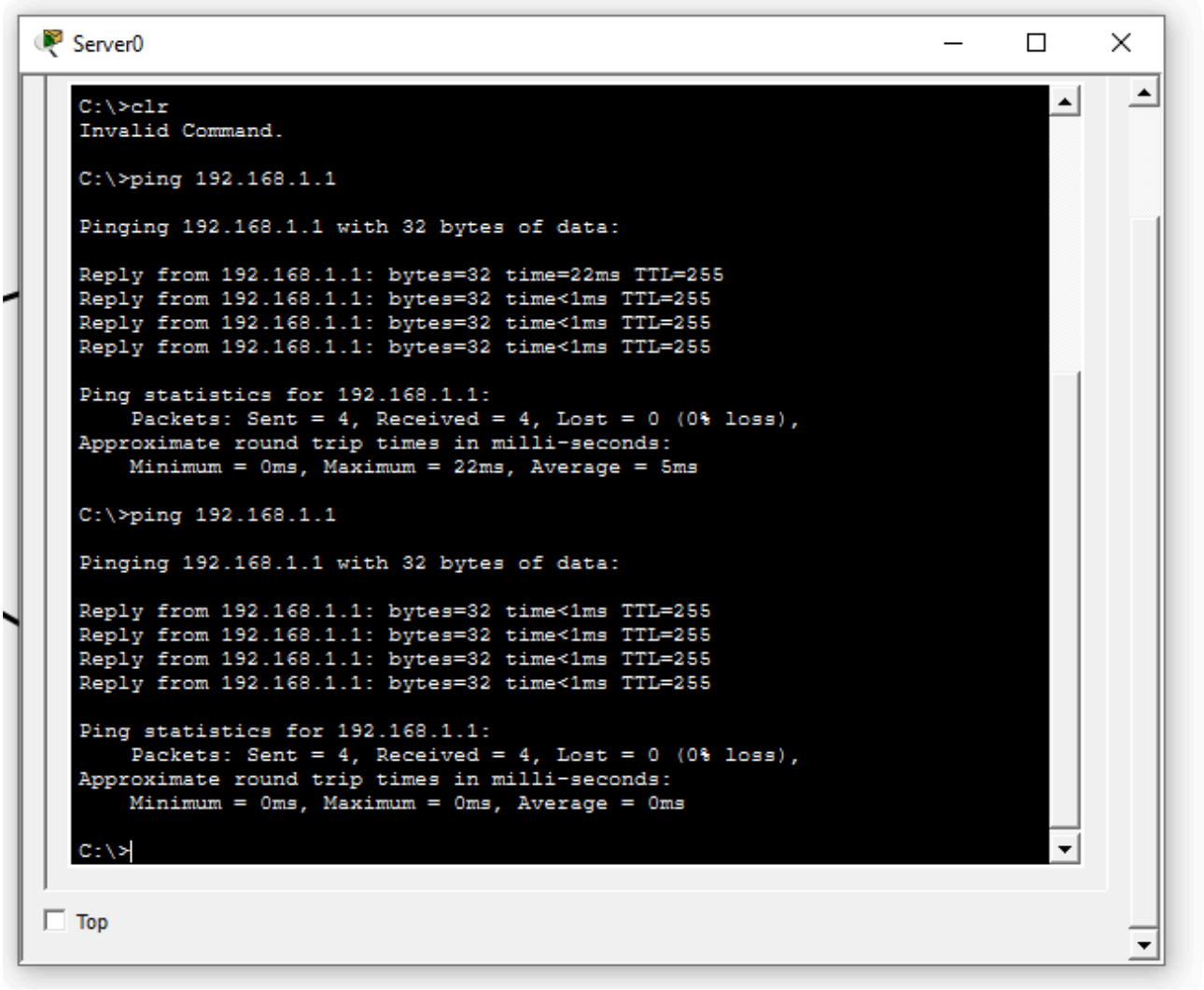


### PC to Server Communication

- PC successfully communicated with the server.
  Shown in the PC1 screen of the above image

### Server to Router Communication

- Server successfully pinged the router with 0% packet loss.

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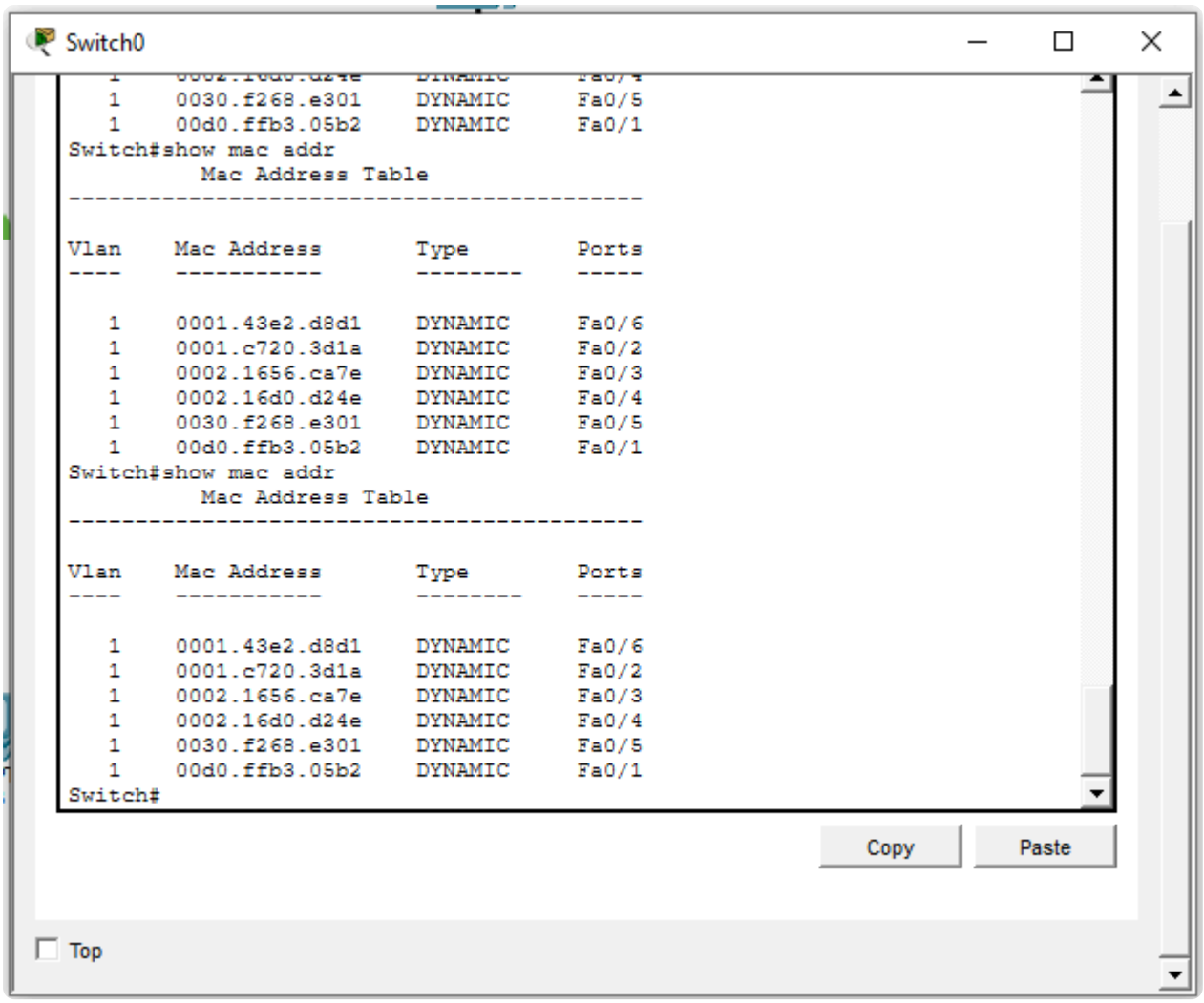
## 7. Switch MAC Address Table Verification

The MAC address table of the switch was checked using CLI:

- Command used:

```
show mac addr
```




- All connected devices (PCs, server, router) were dynamically learned by the switch.

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## Result

- DHCP server successfully assigned IP addresses dynamically to all PCs.
- All devices in the network were able to communicate without any packet loss.
- Router correctly functioned as the default gateway.
- End-to-end connectivity was verified using ping commands.

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## Conclusion

The LAN was successfully implemented using Cisco Packet Tracer with dynamic IP assignment through a DHCP server. All PCs obtained valid IP configurations automatically and communication across the network was verified. Additionally, the switch successfully learned and stored all device MAC addresses in its MAC address table, confirming proper Layer 2 functionality and complete network operation.

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